



MATHIA

LINEARNA ALGEBRA

Formule — Sve formule, teoreme i metode

I deo

Matrice, determinante i rang

II deo

Sistemi linearnih jednačina

III deo

Vektorski prostori

IV deo

Linearne transformacije

V deo

Sopstvene vrednosti i dijagonalizacija

VI deo

Skalarni proizvod i ortogonalnost

Uči s ljubavlju

Tvoja profesorka

I deo Matrice, determinante i rang

Operacije sa matricama

$$(AB)_{ij} = \sum_k a_{ik} b_{kj} \quad (AB)^T = B^T A^T \quad (AB)^{-1} = B^{-1} A^{-1} \quad A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

Determinanta

$$\det A = \sum_j (-1)^{i+j} a_{ij} M_{ij} \quad \det(AB) = \det A \det B \quad \det(A^T) = \det A \quad \det(\alpha A) = \alpha^n \det A$$

Rang

$$\operatorname{rang} A = \dim(\text{prostor kolona}) = \dim(\text{prostor vrsta}); \quad A^{-1} \text{ postoji} \iff \det A \neq 0$$

II deo Sistemi linearnih jednačina

Gaus i Kramer

$$A\vec{x} = \vec{b} \quad x_i = \frac{\det A_i}{\det A} \quad (\det A \neq 0)$$

Kronecker–Capelli

$$\text{saglasan} \iff \operatorname{rang} A = \operatorname{rang} [A \mid b] = r; \quad r = n \Rightarrow \text{jedinstveno}, \quad r < n \Rightarrow \infty$$

III deo Vektorski prostori

Nezavisnost, baza, dimenzija

$$\sum_i \lambda_i v_i = 0 \Rightarrow \lambda_i = 0 \quad \dim V = \# \text{ baze} \quad [v]_B = (\lambda_1, \dots, \lambda_n)$$

Potprostor i promena baze

$$U \leq V : \text{zatvoren za } + \text{ i skalar} \quad [v]_{B'} = P^{-1}[v]_B$$

IV deo Linearne transformacije

Matrica, jezgro i slika

$$f(\alpha u + \beta v) = \alpha f(u) + \beta f(v) \quad M_f = [f(e_1) \cdots f(e_n)]$$

Teorema o rangui i defektu

$$\dim(\operatorname{im} f) + \dim(\ker f) = \dim V \quad A' = P^{-1}AP$$

V deo Sopstvene vrednosti i dijagonalizacija

Sopstveni par

$$\det(A - \lambda I) = 0 \quad (A - \lambda I)v = 0, \quad v \neq 0$$

Dijagonalizacija

$$A = PDP^{-1}, \quad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad P = [v_1 \cdots v_n]$$

VI deo Skalarni proizvod i ortogonalnost

Norma i ortogonalnost

$$\langle u, v \rangle = \sum_i u_i v_i \quad \|u\| = \sqrt{\langle u, u \rangle} \quad u \perp v \iff \langle u, v \rangle = 0$$

Gram–Šmit i ortogonalne matrice

$$w_k = v_k - \sum_{j < k} \frac{\langle v_k, w_j \rangle}{\langle w_j, w_j \rangle} w_j \quad Q^T Q = I \iff Q^{-1} = Q^T$$